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**Key Points:**

- We propose a mixed-flux-based nodal discontinuous Galerkin (DG) method to reduce dynamic rupture simulation dependence on mesh quality
- Our method can handle various complexities in dynamic rupture simulations, as validated with published benchmark problem solutions
- Application of the MixF-DG method to the 2008 Wenchuan earthquake highlights the first-order control of fault geometry on dynamic rupture

**Supporting Information:**

Supporting Information may be found in the online version of this article.

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# A Mixed-Flux-Based Nodal Discontinuous Galerkin Method for 3D Dynamic Rupture Modeling

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**Abstract** Numerical simulation of rupture dynamics provides critical insights for understanding earthquake physics, while the complex geometry and heterogeneous material properties of natural faults make numerical method development challenging. The discontinuous Galerkin (DG) method is suitable for handling these complexities. In the DG method, the fault boundary conditions can be conveniently imposed through the upwind flux by solving a Riemann problem based on a velocity-strain elastodynamic equation. However, the universal adoption of upwind flux can cause spatial oscillations in cases where elements on adjacent sides of the fault surface are not nearly symmetric. Here we propose a nodal DG method with an upwind/central mixed-flux scheme to solve the spatial oscillation problem and thus reduce the dependence on mesh quality. Our method can achieve high scalability in parallel computing under different orders of accuracy. We verify the new method by comparing simulation results with those from other methods on a series of published benchmark problems with complex fault geometries, heterogeneous materials, off-fault plasticity, roughness, thermal pressurization, and various versions of fault friction laws. Finally, we demonstrate that our method can be applied to simulate the dynamic rupture process of the 2008 Mw 7.9 Wenchuan earthquake along realistic fault geometry, including branches, step-overs and a previously neglected sub-parallel fault segment, offering a high potential to systematically explore the influences of fault zone properties on earthquake rupture dynamics.

**Plain Language Summary** Numerical modeling of the earthquake rupture process helps us better understand and investigate the underlying physics of earthquakes. However, it remains challenging to model the fault rupture process for natural earthquakes, partially due to the geometric or/and geological complexities on/around the ruptured faults. To address these complexities, we develop a new numerical method for modeling the 3D fault rupture process of natural earthquakes. In this study, we propose an improved, more flexible numerical scheme to reduce the dependency on mesh quality for earthquake rupture modeling and accommodate complex fault zone properties. We verify the correctness and efficiency of our method by benchmarking several typical models with complex fault geometries (e.g., branch faults and rough faults). We also apply our method to simulate the multi-fault rupture process of the 2008 Wenchuan earthquake to demonstrate the broad potential for natural earthquake modeling applications.

## 1. Introduction

In recent decades, numerical simulation of rupture dynamics has become a powerful means to study the underlying physics of earthquake source mechanisms. Seismogenic faults of natural earthquakes usually exhibit complex fault geometries such as dips, bends, branches, step-overs, and multi-fault coupling. For example, field investigations show the 2008 Wenchuan earthquake ruptured simultaneously on the two imbricate structures of the Beichuan and Pengguan fault (Xu et al., 2009). Seismic and geodetic data joint inversion illustrates the simultaneous ruptures on the plate interface and the overlaying splay faults on the 2016 Kaikōura earthquake (Wang et al., 2018). In addition, the medium around the fault also exhibits strong heterogeneity, which affects the earthquake rupture process. Ulrich et al. (2022) shows the importance of regional-scale structural heterogeneity to the hazards of the 2004 Sumatra–Andaman earthquake and Indian Ocean tsunami. These physical complexities place extremely high demands on the flexibility of numerical methods for rupture dynamics.

Earlier models of earthquake dynamic rupture use the semi-analytical boundary integral equation method (BIEM) (Das, 1976; Das & Aki, 1977). BIEMs are flexible in modeling geometric complex faults (Ando & Kaneko, 2018; Qian et al., 2019) but are limited to homogeneous media. Therefore, numerical methods such as finite difference

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methods (FDMs) (Day, 1982; Madariaga, 1976), finite volume methods (FVMs) (Benjemaa et al., 2009) and finite element methods (FEMs) (Aagaard et al., 2013; Barall, 2009) are applied to three-dimensional (3D) dynamic rupture modelings. The FDM, especially the staggered-grid FDM (Dalguer & Day, 2007; Day et al., 2005), has a simple numerical scheme and high computational efficiency and is one of the most widely used methods. To overcome the limitations of these traditional FDMs on modeling geometrically complex faults, curvilinear FDMs (Kozdon et al., 2012; Z. Zhang et al., 2014; W. Zhang et al., 2020) are developed, which can handle more complex faults such as non-planar faults. Combined with multi-block grid technology, FDMs have been applied to megathrust earthquake ruptures on geometric complex faults such as the 2011 Tohoku-Oki earthquake (Duru & Dunham, 2016; Kozdon & Dunham, 2013; Kozdon et al., 2013).

Despite the high efficiency of FDMs and hexahedral-mesh-based FEMs (Duru et al., 2021; Ely et al., 2009; Galvez et al., 2014; Kozdon et al., 2013; Z. Zhang et al., 2014), it remains challenging to simulate dynamic ruptures on geologically and geometrically complex faults. In contrast, the meshing process for tetrahedral meshes is automatic and more user-friendly even with the open-source meshing software (Geuzaine & Remacle, 2009; Si, 2015). Therefore, tetrahedral mesh-based numerical methods for 3D rupture dynamics are developed, including FEMs (Aagaard et al., 2013; Ma & Archuleta, 2006; Oglesby et al., 2000), FVMs (Benjemaa et al., 2009) and discontinuous Galerkin (DG) methods (also called discontinuous FEM) (Pelties et al., 2012; Tago et al., 2012; Ye et al., 2020). In particular, the DG method combines the high-order advantages of the FEM with the advantages of the easy parallelization of the FVM to become a competitive method for rupture dynamics. In DG methods, various boundary conditions are implemented through numerical fluxes (Cockburn et al., 2012; Hesthaven & Warburton, 2008). Among the different types of numerical fluxes for implementing boundary conditions, the upwind flux (de la Puente et al., 2009; Pelties et al., 2012) can inherently suppress spurious high-frequency oscillations, which is usually achieved by adding artificial viscosity in traditional FEMs (e.g., Aagaard et al., 2013; Galvez et al., 2014).

In this work, we develop a new method for 3D dynamic rupture modelings based on the Gauss-Lobatto-Legendre-based nodal discontinuous Galerkin (NDG) framework (Hesthaven & Warburton, 2008) and apply it to model natural earthquakes with multiple fault segments. We use the velocity-strain form of the elastodynamic equation, which can better describe multi-geophysical problems such as acoustic-elastic coupling under the unified framework (Wilcox et al., 2010; Ye et al., 2016). For the first time, we derive an upwind flux formulation in the form of a velocity-strain equation to impose fault boundary conditions for dynamic rupture problems. The framework of the upwind-flux-based velocity-strain equation has been extended to model seismic waves in more complex cases (e.g., anisotropy, viscoelasticity) (Zhan et al., 2020). Therefore, the use of the velocity-strain equation facilitates the incorporation of these complexities into dynamic rupture problems. The NDG method we use here is mathematically equivalent to the modal DG method but different in terms of numerical implementation (Hesthaven & Warburton, 2008). Extra conversions of the modal and nodal coefficients are required in the modal DG framework for the implementation of Drucker–Prager viscoplasticity (Wollherr et al., 2018). In contrast, viscoplasticity can be straightforwardly incorporated into the NDG framework.

The upwind flux is advantageous in suppressing spurious high-frequency oscillations (HFOs), resulting in smooth and accurate simulated time series of the rupture process (de la Puente et al., 2009). However, in some dynamic rupture modeling cases, the universal adoption of upwind flux for all boundary conditions can be problematic. If the mesh adjacent to the fault is not approximately symmetric, the use of upwind flux can lead to the results with spatial spike oscillations (SSOs), especially in the normal or shear stress components. The SSOs are also present in the modal DG method using upwind flux (Breuer & Cui, 2016, 2018) and is not a bug in the code implementation. A detailed comparison of modeling results with the regular, frontal and unstructured Delaunay meshing strategies shows that instabilities of SSOs are strongly mesh-dependent (Breuer & Cui, 2016), which are not resolved in their later attempts using a posterior sub-cell limiter (Breuer & Cui, 2018). The existence of SSOs makes mesh generation for dynamic rupture problems challenging. For some simple geometries, such as a vertical planar fault, we can construct a mirrored mesh to make sure the mesh is symmetric to the fault surface, resulting in oscillation-free simulation results. While for non-planar faults or low-dip-angle thrust faults, it is impossible to generate a perfectly symmetric mesh and challenging to generate a nearly symmetric mesh. To solve the problem of SSOs, we propose a mixed-flux scheme, that is, using a mixture of upwind and central fluxes for different types of boundaries. The benchmark examples we tested show that the adoption of the mixed-flux schedule removes the mesh-induced SSOs in the upwind-flux-based DG method, thus reducing the numerical dependence on mesh quality, a critical challenge to simulate rupture dynamics on complex fault geometries.

We tested our NDG method with MPI and found good parallel scalability for tests up to about 500 CPU processors. At the current stage, the parallel scalability of this method has enabled us to simulate the rupture dynamics of various complex fault systems of large earthquakes, and our method holds the potential to adapt to a larger parallel scale in supercomputers (Fu et al., 2017). By comparing with benchmark examples from the *SCEC/USGS Spontaneous Rupture Code Verification Project* (Harris et al., 2009, 2018), we verify the accuracy and flexibility of our method in modeling rupture dynamics with the bimaterial property, branching faults, off-fault plasticity, fault roughness, thermal pressurization and various friction laws. Finally, we apply our new method to simulate the dynamic rupture process of the 2008 Wenchuan earthquake, including topography and complex geometries with multi-faults in our simulations. In particular, the flexibility of our method allows the inclusion of the Xiaoyudong fault, a ∼7 km long segment approximately perpendicular to and connecting the sub-parallel Beichuan and Pengguan faults. The Xiaoyudong fault is documented to have experienced coseismic rupture during the 2008 Wenchuan earthquake (Tan et al., 2012; Xu et al., 2009), but was neglected in previous dynamic rupture simulations due to numerical challenges (Tang et al., 2021). Both benchmark models and the Wenchuan earthquake model illustrate the advantages of our new method in simulating the dynamic rupture process on complex fault systems.

## 2. The Nodal Discontinuous Galerkin Method

In this section, we explain the framework of the NDG method and demonstrate how to implement fault boundary conditions by solving the exact Riemann problems under the velocity-strain form of elastodynamic equations.

### 2.1. DG Discretization

The general form of the governing equations for an elastodynamic problem consists of the momentum equation:

$$\rho \frac{\partial^2 \boldsymbol{u}}{\partial t^2} = \nabla \cdot \boldsymbol{\tau}, \tag{1}$$

and the constitutive equation:

$$\boldsymbol{\tau} = \boldsymbol{c} : : \boldsymbol{\varepsilon}, \tag{2}$$

where $\boldsymbol{u}$ is the displacement vector and $\rho$ is the density of the material. $\boldsymbol{\tau}$ and $\boldsymbol{\varepsilon}$ are the stress and strain tensors, related by the elastic modulus tensor $\boldsymbol{c}$ (Aki & Richards, 2009). A first-order velocity-strain form of elastodynamic equations is adopted here, which is suitable to describe multi-physics problems under the same unified framework (Wilcox et al., 2010; Ye et al., 2016; Zhan et al., 2020). We consider isotropic elastic media in this work, yielding governing equations of the form:

$$\frac{\partial \boldsymbol{q}}{\partial t} = \frac{\partial \boldsymbol{f}(\boldsymbol{q})}{\partial x} + \frac{\partial \boldsymbol{g}(\boldsymbol{q})}{\partial y} + \frac{\partial \boldsymbol{h}(\boldsymbol{q})}{\partial z}, \tag{3}$$

where the solution vector $\boldsymbol{q}$ consists of velocity and strain variables:

$$\boldsymbol{q} = (\rho v_x, \rho v_y, \rho v_y, \varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, \gamma_{yz}, \gamma_{xz}, \gamma_{xy})^T, \tag{4}$$

where $\rho$ is density, $v_i$ is the particle velocity, $\varepsilon_{ij}$ is the strain and $\gamma_{ij}$ is the engineering strain ($\gamma_{ij} = 2\varepsilon_{ij}$). $\boldsymbol{f}, \boldsymbol{g}, \boldsymbol{h}$ are flux functions in $x$, $y$, $z$ directions:

$$\boldsymbol{f} = (\tau_{xx}, \tau_{xy}, \tau_{xz}, v_x, 0, 0, 0, v_z, v_y)^T, \tag{5a}$$

$$\boldsymbol{g} = (\tau_{xy}, \tau_{yy}, \tau_{yz}, 0, v_y, 0, v_z, 0, v_x)^T, \tag{5b}$$

$$\boldsymbol{h} = (\tau_{xz}, \tau_{yz}, \tau_{zz}, 0, 0, v_z, v_y, v_x, 0)^T, \tag{5c}$$

where $\tau_{ij}$ are the components of the stress tensor $\boldsymbol{\tau}$.

Let $\Omega$ denote the computational domain (Figure 1). $\Omega$ is discretized into $N_e$ non-overlapping tetrahedral elements $\Omega_i (i = 1, 2, \ldots, N_e)$ with the boundary $\partial\Omega_i$. The solution vector in each element $\Omega_i$ is approximated as:

$$q_i(r,t) = \sum_{k=1}^{N_p} q_{i,k}(t) l_k(r), \tag{6}$$

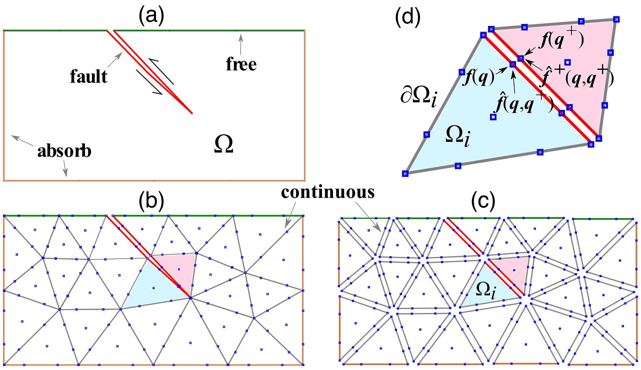


**Figure 1.** (a) A 2D schematic dynamic rupture model, Ω denotes the entire computational domain, including the fault (red lines), free (green lines), and absorbing boundaries (magenta lines). (b) The continuous Galerkin (traditional FEM) discretization, solution vectors $\boldsymbol{q}$ are defined in the nodal points (blue squares). Fault surfaces are represented as "split nodes" (points share locations but have double values), and continuous boundaries (gray lines) are represented as shared points. (c) In the discontinuous Galerkin (DG) discretization, both fault and continuous boundaries are represented as "split nodes." (d) The element $\Omega_i$ with three boundaries $\partial\Omega_i$ and one of its neighboring elements. $\boldsymbol{f}(\boldsymbol{q}), \boldsymbol{f}(\boldsymbol{q}^+)$ are the flux function vectors. $\hat{\boldsymbol{f}}(\boldsymbol{q},\boldsymbol{q}^+)$, $\hat{\boldsymbol{f}}^+(\boldsymbol{q},\boldsymbol{q}^+)$ are the "numerical" flux vectors, which are used to implement all boundary conditions in the DG framework.

here $l_k$ is the nodal Lagrangian basis function with the maximum number of $N_p = \frac{N(N+1)(N+2)}{6}$, $N$ is the polynomial order. Therefore, the expansion coefficients $q_{i,k}$ are also the nodal solution variables on the collocation points $r$ on $\Omega_i$. Following the classic workflow of the DG method (Texts S1 and S2 in Supporting Information S1), we apply the DG testing (set test function as basis function) to the velocity-strain form of elastodynamic equations:

$$\int_{\Omega_i} l_k[\partial_t \boldsymbol{q} - \nabla\cdot(\boldsymbol{f},\boldsymbol{g},\boldsymbol{h})]\mathrm{d}V = \int_{\partial\Omega_i} l_k T^{-1}\left(\hat{\boldsymbol{f}} - \boldsymbol{f}\right)\mathrm{d}S, \tag{7}$$

where $\hat{\boldsymbol{f}}$ is the "numerical" flux, which is a combination of the solution vectors in the split faces of the two adjacent elements: $\hat{\boldsymbol{f}} = \hat{\boldsymbol{f}}(\boldsymbol{q},\boldsymbol{q}^+)$ (see Figure 1, the superscript "+" indicates the neighboring elements). Numerical flux is the core of the FVM and DG methodologies (LeVeque, 2002; Toro, 2009). $T$ is the rotation matrix that is used to rotate the global coordinate to the element face-aligned local coordinate (Text S3 in Supporting Information S1) for the convenience of deriving the numerical flux. Note that the flux $\boldsymbol{f}, \hat{\boldsymbol{f}}$ in the right term of Equation 7 is in the local coordinate. Once the numerical flux $\hat{\boldsymbol{f}}$ is derived in the local coordinate, it should be rotated back into the global coordinate using the inverse rotation matrix $T^{-1}$.

### 2.2. Boundary Conditions

How to implement the boundary conditions is crucial for developing numerical methods for dynamic rupture models. There are four types of boundary conditions in the model (Figure 1): (a) fault surface, where the elements have relative motion (dislocation); (b) free surface, where the traction force vanishes; (c) absorbing boundaries of the model domain, where inward-going waves vanish; (d) continuous boundaries, unbroken elements without relative motion. In the continuous Galerkin (traditional FEM) framework, continuous boundary conditions are implicitly implemented as the solution/flux vectors in the boundary share the same value: $\boldsymbol{q} = \boldsymbol{q}^+, \boldsymbol{f}(\boldsymbol{q}) = \boldsymbol{f}(\boldsymbol{q}^+)$. While in the DG framework, all interior boundaries (including continuous boundaries) are represented as split nodes with double values, that is, $\boldsymbol{f} \neq \boldsymbol{f}^+$ holds even for continuous boundary conditions (Figure 1d). All the boundary conditions are implemented in the numerical flux term (e.g., $\hat{\boldsymbol{f}} = \hat{\boldsymbol{f}}^+$ for continuous boundary conditions), rather than the flux term $\boldsymbol{f}$.

In this work, we choose the upwind flux for the numerical flux term $\hat{\boldsymbol{f}}$ (Equation 7) for its advantage for suppressing spurious high-frequency oscillations (de la Puente et al., 2009). However, in Section 3, we show that the universal adoption of the upwind flux for all boundary conditions can be problematic in some dynamic rupture cases and a modification of upwind flux will be introduced. In the following two subsections, we start to derive the upwind flux for a dynamic rupture model by introducing the Riemann solvers.

#### 2.2.1. Riemann Solver for Continuous Boundaries

Following the framework of Toro (2009) and Zhan et al. (2020), assuming $x$ is the normal direction of the element boundary, we solve the Riemann problem under the Rankine-Hugoniot conditions (Text S4.1 in Supporting Information S1) for the continuous boundary condition $\left(\hat{\boldsymbol{f}}^+ = \hat{\boldsymbol{f}}\right)$:

$$[\![\hat{v}_\theta]\!] = 0, [\![\hat{\tau}_{x\theta}]\!] = 0, (\theta = x, y, z), \tag{8}$$

yielding the upwind flux:

$$\hat{\boldsymbol{f}} - \boldsymbol{f} = (Z_p\alpha_1, Z_s\alpha_2, Z_s\alpha_3, \alpha_1, 0, 0, 0, \alpha_3, \alpha_2)^T, \tag{9}$$

where $\alpha_1 = \frac{[\![\tau_{xx}]\!] + Z_p^+[\![v_x]\!]}{Z_p + Z_p^+}$, $\alpha_2 = \frac{[\![\tau_{xy}]\!] + Z_s^+[\![v_y]\!]}{Z_s + Z_s^+}$, $\alpha_3 = \frac{[\![\tau_{xz}]\!] + Z_s^+[\![v_z]\!]}{Z_s + Z_s^+}$, $Z_p$, $Z_s$ are the impedance of P and S waves. The superscript "+" indicates the neighboring elements. $[\![\cdot]\!]$ denotes the difference between the solutions on the face of two adjacent elements: $[\![\theta]\!] \equiv \theta^+ - \theta$. For the free surface boundaries and the exterior boundaries, Equation 9 is also used by setting the artificial solution vector $\boldsymbol{q}^+$ as stress imaged solution $\left(\tau^+_{x\theta} = -\tau_{x\theta}, \theta = x, y, z\right)$ or vanished solution ($\boldsymbol{q}^+ = 0$), respectively.

### 2.2.2. Riemann Solver for Dynamic Ruptured Boundaries

It is straightforward to derive the Riemann solver of ruptured boundaries following the same framework (Toro, 2009; Zhan et al., 2020) for continuous boundaries. The spontaneous rupture problem requires mixed continuous-discontinuous boundary conditions, that is, $\hat{\boldsymbol{f}}^+ \neq \hat{\boldsymbol{f}}$ (the stress components are continuous while the velocities are discontinuous). We rotate the coordinate to set the fault normal direction as $x$, hence the fault slips at $y$ and $z$ direction. Then the fault boundary conditions are expressed using the numerical flux as follows:

$$[\![\hat{v}_x]\!] = 0, [\![\hat{v}_y]\!] = V_y, [\![\hat{v}_z]\!] = V_z, [\![\hat{\tau}_{x\theta}]\!] = 0, (\theta = x, y, z). \tag{10}$$

Solving the Riemann problem for dynamic rupture boundary conditions (Text S4.2 in Supporting Information S1) yields the following relationship:

$$\hat{\tau}_{x\theta} = \Phi_\theta - \eta V_\theta, (\theta = y, z), \tag{11}$$

where $\eta = \frac{Z_s Z_s^+}{Z_s + Z_s^+}$. $\Phi_\theta$ is the shear stress component when the fault is locked (Equation 12, $\Phi_y = Z_s\alpha_2$, $\Phi_z = Z_s\alpha_3$). Applying the parallel condition $\frac{\tau}{|\tau|} = \frac{\boldsymbol{V}}{|\boldsymbol{V}|}$, we obtain the relationship for the absolute slip rate $V$ and shear stress $\tau$:

$$\hat{\tau} = \Phi - \eta V. \tag{12}$$

Equation 12 is crucial for implementing fault boundary conditions. Combining Equation 12 and the friction laws, we can solve the stress shear and the slip rate on the fault. Next, we will illustrate the details for implementing slip-weakening and rate-state friction laws.

For slip-weakening friction law, the friction coefficient is slip dependent:

$$\mu_f = \mu_f(s) = \max\left\{\mu_d, (\mu_s - \mu_d)\frac{s}{D_c}\right\}, \tag{13}$$

where $\mu_s$, $\mu_d$ is the static and dynamic friction, $D_c$ is the characteristic slip distance. $s$ is the fault slip which can be obtained by integrating fault slip rate $V$. The fault normal stress is solved by implementing the continuous and the "non-open" condition:

$$\sigma = \min\left\{0, \hat{\tau}_{xx} + \tau^0_{xx}\right\}, \tag{14}$$

the superscript "0" in $\tau^0_{xx}$ denotes prestress. $\hat{\tau}_{xx}$ is the numerical flux of the normal stress component when the fault is locked. From Equation 9 we can know $\hat{\tau}_{xx} = Z_p\alpha_1$. If $\hat{\tau} = \sqrt{\hat{\tau}^2_{xy} + \hat{\tau}^2_{xz}}$ is larger than the shear strength $\mu_f\sigma$, which means that the fault cannot remain locked and begins to slip, then the shear stress should be $\mu_f\sigma$. Otherwise, when the fault stress $\tau$ is below the level of the fault strength, indicating the fault is locked, the absolute fault shear stress should be $\hat{\tau} + \hat{\tau}^0$. The formula of relative fault stress under these two circumstances can be summarized as the following equation:

$$\hat{\tau} = \min\left\{\hat{\tau}, \mu_f\sigma - \tau^0\right\}, (\theta = y, z). \tag{15}$$

Once we obtain the shear stress, the slip rate can be solved by the relationship between shear stress and slip rate (Equation 12).

For rate-state friction laws, since the fault is always sliding, then we can directly set the shear stress as $\mu_f\sigma$. Combined with Equation 12, a nonlinear equation can be obtained:

$$\hat{\tau} = \Phi - \eta V = \mu_f(V, \psi)\sigma. \tag{16}$$

In this case, the friction $\mu_f$ is rate ($V$) and state ($\psi$) dependent. A nonlinear solver, such as Newton-Rasphson or Regula Fasi, can be used to solve the slip rate $V$ in Equation 16. The state variable $\psi$, obeying different evolution laws: $\frac{\mathrm{d}\psi}{\mathrm{d}t} = G(V, \psi)$, is updated by the same time integration method for solution vector $\boldsymbol{q}$.

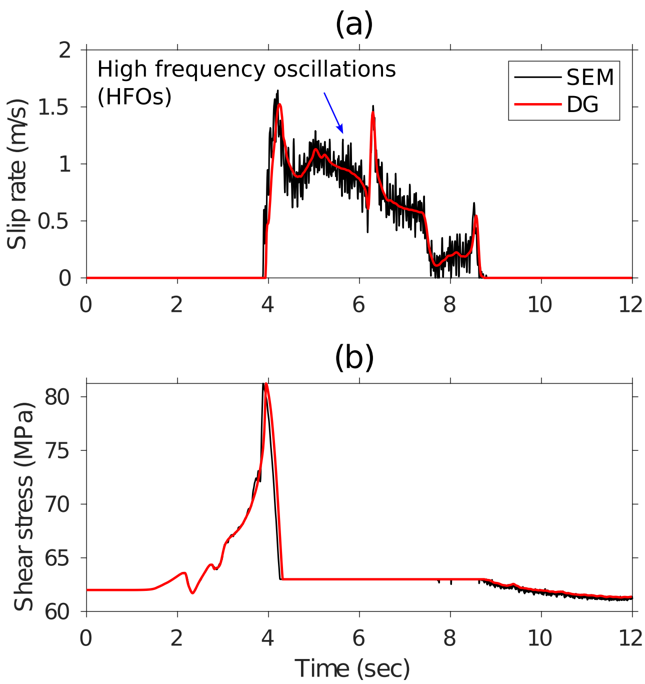


**Figure 2.** An example showing the spurious high-frequency oscillations (HFOs) using the benchmark model TPV5 (Harris et al., 2009). The solution data is downloaded from the benchmark website (https://strike.scec.org/cvws) and plotted here. (a) Slip rate (b) shear stress time series on a station of the fault surface, which is 7.5 km away from the nucleation center along the strike, and at the same depth as the nucleation center. The black color indicates the results of the SEM (Kaneko et al., 2008) and the red color is calculated by a modal DG method (Pelties et al., 2012).

### 2.3. Time Discretization

In this work, we adopt the fourth-order low-storage explicit Runge-Kutta method (LSERK) (Carpenter & Kennedy, 1994) for the time integration. Discussions on LSERK can be found in many works (Cockburn & Shu, 2001; Hesthaven & Warburton, 2008). The time step is restricted by the Courant–Friedrichs–Lewy (CFL) number (Courant et al., 1928):

$$\Delta t \leq C \frac{\min_j \Delta x_j}{\max_j V_{p_j}}, \tag{17}$$

where $C$ is the CFL number, $\Delta x_j$ and $V_{p_j}$ are the grid size and P-wave velocity of cell $j$. CFL number is dependent on the order and mesh quality (Cockburn & Shu, 2001). In practice, we set the time step to 50% of the value given by Equation 17 for safety.

## 3. Method Improvement

In this section, we will demonstrate that the upwind flux introduced in the previous section is problematic when the near-fault mesh is not nearly symmetric in some simulation cases. We then propose a new mixed flux scheme to improve the performance of the DG method for dynamic rupture modeling.

### 3.1. Upwind Flux in Rupture Dynamics: A Double-Edged Sword

In this section, we introduce the advantage of upwind flux: suppression of spurious high-frequency oscillations (HFOs), and its disadvantage: possible spurious spatial spike oscillations (SSOs). Note that although both HFOs and SSOs are oscillations, they are two completely different numerical artifacts, precisely the two sides of the double-edged sword of upwind flux in dynamic rupture modeling.

#### 3.1.1. Advantage: Suppression of Spurious High-Frequency Oscillations (HFOs)

The upwind-flux-based DG methods are naturally dissipative and free of spurious high-frequency oscillations and are therefore intensively used in wave propagation problems (Käser & Dumbser, 2006; Wilcox et al., 2010; Zhan et al., 2020) and dynamic rupture modelings (de la Puente et al., 2009; Pelties et al., 2012). Figure 2 shows an example of different oscillation behaviors of the upwind-flux-based DG method (Pelties et al., 2012) and the spectral element method (SEM) (Kaneko et al., 2008). Using a non-dissipative central flux have a similar spurious problem of HFOs (Tago et al., 2012, Figure 9 therein) as the SEM. These studies show the use of upwind flux is advantageous for suppressing the spurious HFOs in dynamic rupture modeling compared to non-dissipative methods (central-flux-based DG or SEM).

#### 3.1.2. Disadvantage: Possible Spurious Spatial Spike Oscillations (SSOs)

Despite its advantage of suppressing spurious HFOs, we found that the use of the upwind flux results in other types of numerical "oscillations" under certain circumstances. As shown in Figure 3, when the mesh adjacent to the fault is not symmetric, the use of upwind flux is prone to spatial spike oscillations (SSOs), especially in the stress components (Figures 3d–3f). The same SSO phenomenon is also documented in preliminary reports of other modal DG methods (Breuer & Cui, 2016, 2018) (see Text S10 in Supporting Information S1). The SSOs discussed here are completely different from the HFOs shown in Figure 2. As shown in Figure 3e, the time series (red curve) is very smooth and does not contain any oscillations, although it deviates from the true value (blue curve). The SSOs do not appear in the time series but in space, therefore called spatial spikes oscillations (Figure 3f).

The HFOs (Figure 2) are mesh-independent and can be suppressed by adding artificial damping (e.g., Kelvin-Voigt viscosity). It does not lead to numerical instabilities and can be reduced by mesh refinement. In contrast, the

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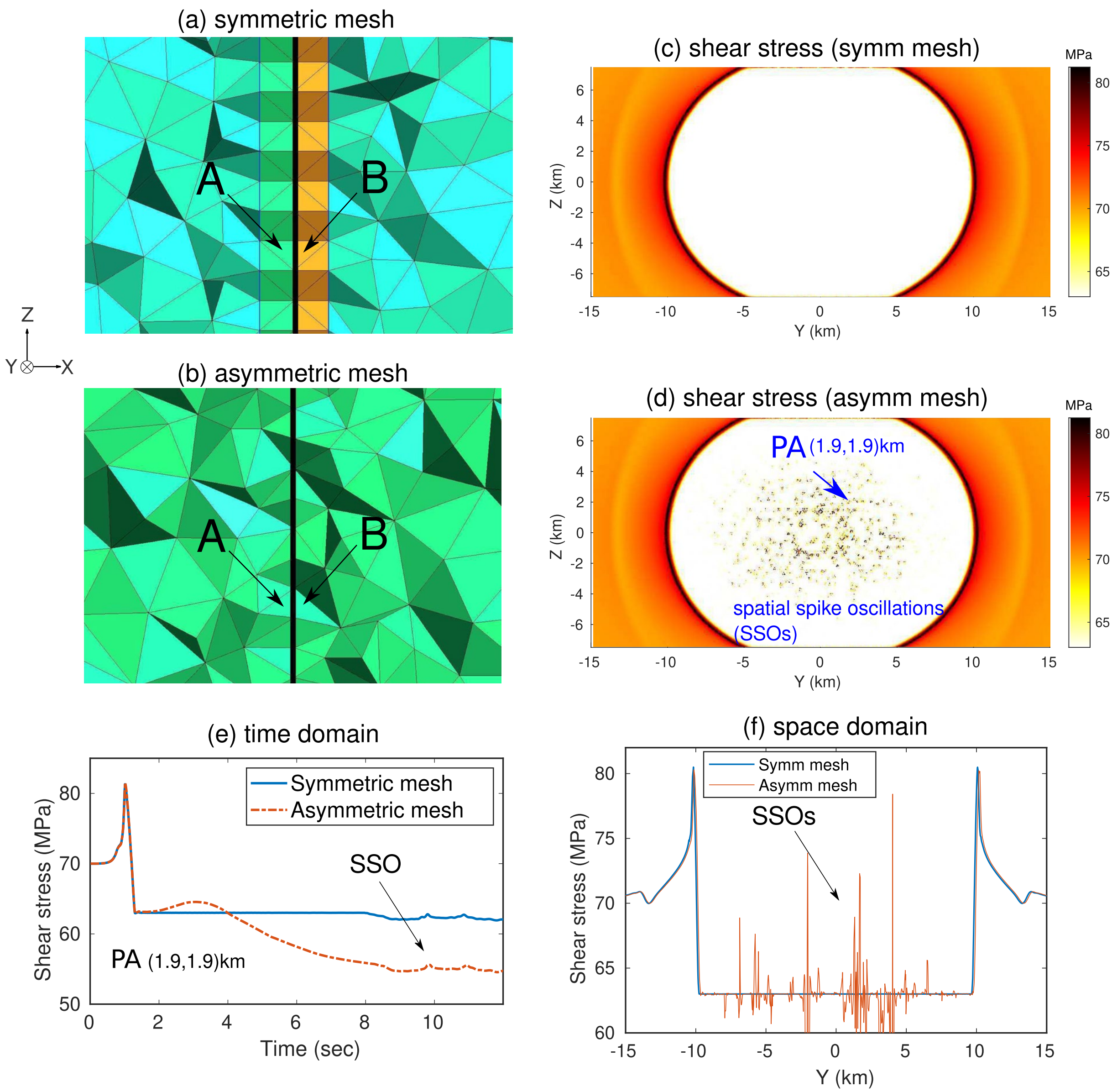


**Figure 3.** An example showing the mesh induced "spatial spike oscillations" (SSOs). (a) and (b) are cross sections (*X–Z* plane) perpendicular to the fault surface (the *Y–Z* plane). (a) Symmetric mesh with an extruded layer; (b) asymmetric mesh without an extruded layer. A and B are two neighboring elements that are immediately adjacent to the fault. Both meshes are generated by the Gmsh software (Geuzaine & Remacle, 2009) and strong mesh coarsening is applied (The edge length of the element size near the fault is 200 m, and increases to 5 km at the domain boundary). The "frontal-Delaunay" algorithm is used to generate the 2D surface mesh and the "Delaunay" algorithm to generate the 3D volumetric mesh. (c) and (d) are the snapshots of shear stress at a time of 4 s calculated with mesh (a) and (b). (e) The "spatial spike" of station PA in the time domain when the asymmetric mesh is used. (f) The "spatial spike" in the space domain (spatial profile of $Z = 0$).

SSOs (Figure 3) are mesh-dependent and can lead to serious instability, especially when the mesh size is refined and/or the polynomial order increases since the "filtering effect" by the dissipation of the upwind flux is less.

We need to emphasize that the instability of SSOs only appears when certain types of meshes are involved. Based on the comparison in Figure 3, we found the asymmetry of the mesh adjacent to the fault is the most important contributing factor. Therefore, we define a parameter to measure the mesh quality for dynamic rupture models:

$$r_v \equiv \max\left\{\frac{V_A}{V_B}, \frac{V_B}{V_A}\right\}, \tag{18}$$

where $A$ and $B$ are the pair of neighboring elements adjacent to the fault surface (Figures 3a and 3b), $V_A$ and $V_B$ are respective volumes and $r_v$ is the volume ratio. When the elements adjacent to the fault are completely symmetric

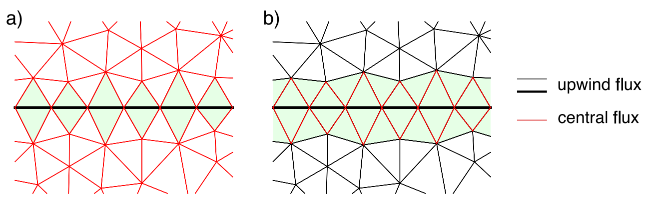


**Figure 4.** A 2D schematic illustration for the scheme of (a) "mixed flux 1" and (b) "mixed flux 2." The fault surface is highlighted by the thick black line. The light green color indicates the elements using mixed fluxes. In each light green element, the upwind flux is used for the black edges (including the fault), while the central flux is used for the red edges. In this 2D example, a total of 3,090 edges require numerical flux to implement boundary conditions. For (a), 2,994 edges (97%) use central flux (red color); for (b), only 250 edges (8%) use central flux (red color).

(Figure 3a), $r_v$ is 1 (note that $r_v = 1$ does not imply symmetry). When $r_v$ is much larger than 1, it indicates that the elements adjacent to the fault are very asymmetric (Figure 3b), which leads to SSOs shown in Figure 3d. While a perfectly symmetric mesh is ideal, for many non-planar faults it is very difficult to achieve. In our experience, when $r_v > 1.5$, SSOs like Figure 3d will occur. For 3D dynamic rupture problems, the use of unstructured triangular meshes cannot always guarantee that $r_v < 1.5$, especially when the strong mesh coarsening strategy is implemented. This suggests that the problem of upwind flux in rupture dynamics needs to be addressed.

### 3.2. The Mixed Flux

Here we introduce a "mixed flux" scheme to improve the performance of the upwind flux in dynamic rupture modelings. Mixed flux, as the name suggests, is a mix of upwind flux and other flux (here, the central flux). The application of specific types of numerical flux schemes will be dependent on the types of boundary conditions. Figure 4 is a schematic diagram of our proposed mixed flux method. In the first scenario "mixed flux 1," we use central flux for all the continuous boundary conditions (Equation 9):

$$\hat{\boldsymbol{f}}\left(\boldsymbol{q}, \boldsymbol{q}^{+}\right) = \frac{\boldsymbol{f}(\boldsymbol{q}) + \boldsymbol{f}(\boldsymbol{q}^{+})}{2} \tag{19}$$

while keeping the upwind flux for the rest of the boundaries, including the fault surfaces, free surfaces, and absorbing boundaries (Figure 4a). But since most of the element boundaries are continuous boundaries, "mixed flux 1" (Figure 4a) results in an extremely high proportion (e.g., 97% in the case of Figure 4a) of central flux usage. That is, mixed flux 1 is close to the central flux method, which is prone to spurious high-frequency oscillations (Tago et al., 2012) (note that it is different from the spatial spikes discussed in the previous section). To solve this problem, we propose a more advantageous "mixed flux 2" scheme by limiting the use of the central flux only on the surfaces immediately adjacent to the fault, while for other surfaces the upwind flux is still used to impose continuous boundary conditions (Figure 4b).

To further elaborate on the possible mechanisms of why the upwind flux is problematic when the mesh is highly asymmetrical, we use the schematic diagram (Figure 5) to explain the cause of SSOs. Under the numerical framework of DG, a certain boundary point on the fault surface will be shared by different boundaries and become multiple split nodes (blue dots in Figure 5). These points require different enforcement of boundary conditions depending on which boundary it belongs to. For example, points A and B are located on both sides of the fault boundary $e_{AB}$ (same as $e_{EF}$) where discontinuous boundary conditions need to be applied, whereas points A and C are located across the continuous boundary $e_{AC}$ (same as $e_{BD}$, $e_{DF}$, and $e_{CE}$).

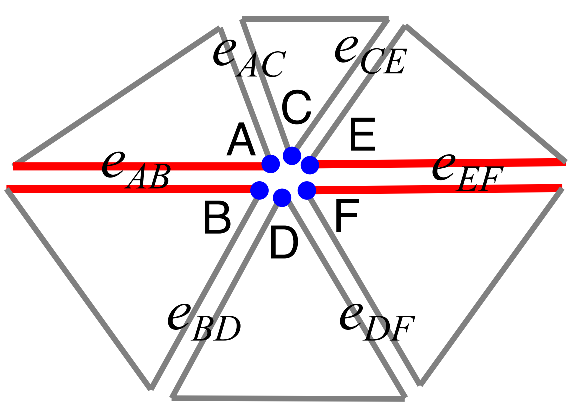


**Figure 5.** Schematic diagram of the mesh near the fault. The red line denotes the fault boundary (jump boundary) while the gray line is the continuous boundary. Points A–F (blue circles) are the split nodes on the fault (they share coordinates but have different values). The edges ($e_{AB}$, $e_{AC}$, …) require the implementation of continuous (gray) or jump (red) boundary conditions.

Previous works (He, Yang, & Qiu, 2020; He et al., 2019) have shown that both the mesh size and mesh shape greatly affect the numerical dispersion and dissipation of the upwind flux. Therefore, if the mesh near the fault is highly asymmetric (e.g., $r_v > 3$), the numerical dissipation of the upwind flux will be very different on the two sides of the fault. The unbalanced dissipation error on the continuous-discontinuous fault boundaries is likely the culprit of the spatial spikes. When the lengths of $e_{AC}$ and $e_{BD}$ differ greatly, their dissipation errors are also quite different, which is prone to numerical problems. One significant nature of dynamic rupture is that when asymmetries occur (e.g., tilting (Oglesby et al., 2000), bi-materials (Cochard & Rice, 2000)), there will be perturbations in normal stresses. The asymmetric dissipation errors caused by upwind flux under asymmetric grids also have similar effects. However, the geometry-induced stress perturbations are artificial and need to be suppressed from divergence. Based on these considerations, if we want to balance the different magnitudes of dissipation errors across $e_{AC}$ and

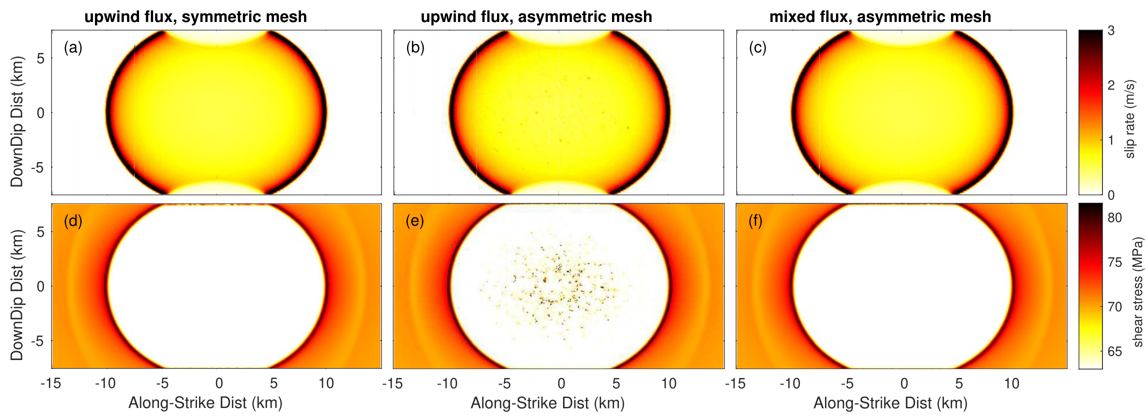


**Figure 6.** Comparison of results calculated by different numerical flux schemes and different meshes. The top and bottom rows show snapshots of slip rate and shear stress, respectively, at 4 s (a), (d): symmetric mesh + upwind flux; (b), (e): asymmetric mesh + upwind flux; (c), (f): asymmetric mesh + mixed flux (Figure 4b, the mixed flux 2).

$e_{BD}$, the most direct way is to use a non-dissipative central flux. That's how we came up with the idea of mixed flux.

### 3.3. Comparison of Upwind and Mixed Flux

We use the following two examples directly and modified from the *SCEC/USGS Spontaneous Rupture Code Verification Project* (Harris et al., 2009, 2018) to demonstrate the mixed flux scheme can greatly reduce the dependence on mesh quality when modeling a dynamic rupture process.

The first example is benchmark problem TPV3, which is a vertical right-lateral strike-slip fault model (30 km × 15 km) in a homogeneous full space, governed by a linear slip-weakening friction law; details of TPV3 setup can be found at the project website https://strike.scec.org/cvws and model parameters are listed in Table S1 of the Supporting Information S1. TPV3 has been intensively used for method verification in previous works (Pelties et al., 2012; Tago et al., 2012; Z. Zhang et al., 2014).

To illustrate the effect of mesh quality on the rupture process, two types of meshes are generated using the Gmsh software (Geuzaine & Remacle, 2009). For the first set of mesh, we add two extra layers, one on each side of the fault surface, to ensure that each pair of elements adjacent to the fault are completely symmetric (Figure 3a). Therefore, the volume ratio $r_v$ is 1 for all the pairs of elements with edges on the fault surface. As a comparison group, the second set of mesh does not include the extra layers on both sides of the fault surface (Figure 3b). Therefore, the volume ratio $r_v$ of many pairs of elements immediately adjacent to the fault surface exceeds 1.5. We use the "frontal-Delaunay" algorithm to generate the 2D surface mesh and the "Delaunay" algorithm to generate the 3D volumetric mesh. Both meshes use strong mesh coarsening. The edge length of the elements near the fault is 200 m and is coarsened to 5 km at the domain boundary. We use the spatial order of $O = 3$ to perform the simulations.

We first calculate the results using upwind flux and a symmetric mesh, as shown in Figures 6a and 6d. The results show that both the slip rate and the shear stress are smooth, with no spatial oscillations (SSOs), suggesting that upwind flux is a suitable choice when the mesh quality is good. By contrast, the application of upwind flux to an asymmetric mesh leads to severe SSOs (Figures 6b and 6e). The SSOs disappear with the mixed flux strategy despite the asymmetric mesh (Figures 6c and 6f), generating slip rate and shear stress distributions very similar to those using the upwind flux with symmetric mesh (Figures 6a and 6d). This shows that our improved mixed flux can achieve satisfactory results even with asymmetric meshes, whereas the upwind flux cannot.

Note that in order to handle the SSOs we introduce a non-dissipative central flux into the mixed-flux scheme, which in theory would result in HFOs (Section 3.1.1) even if the proportion of central flux is low (e.g., 8% in Figure 4). As shown in Figure 6, there is little difference in the results of mixed-flux and upwind-flux (symmetric) with a fine mesh of 200 m. To clearly compare the influence of the central flux on the dissipation, we compared the results of different fluxes under the symmetric (no SSOs would arise) coarse grid in Supporting

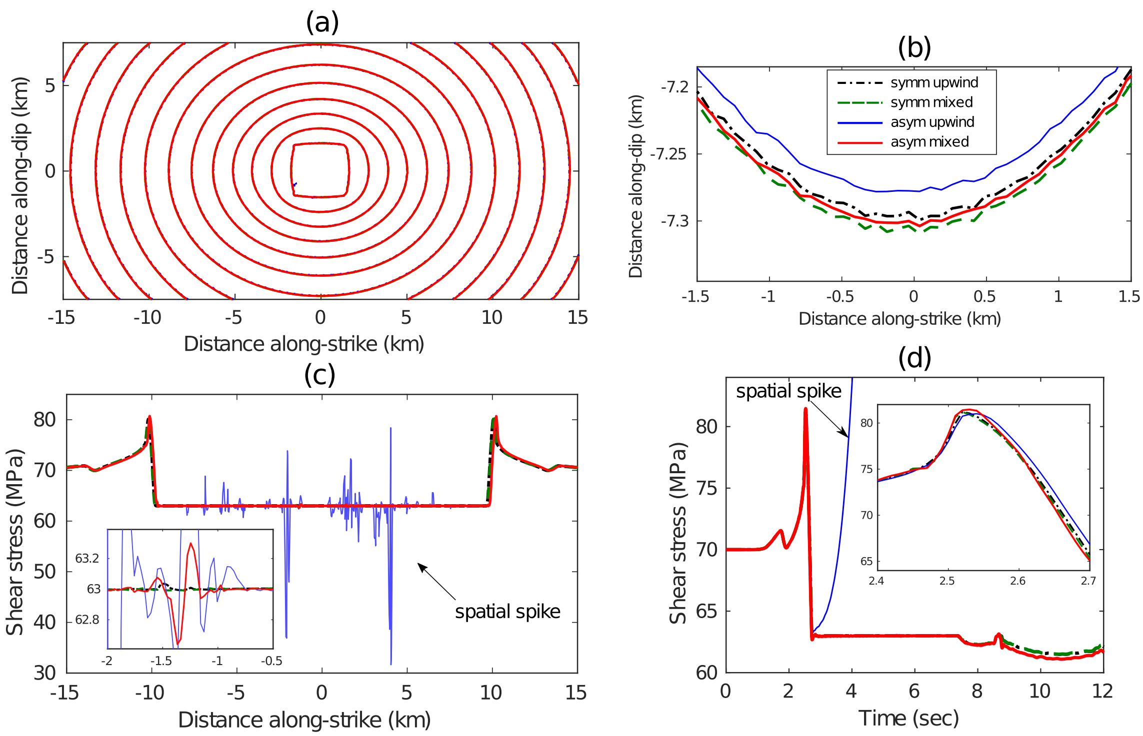


**Figure 7.** (a) Comparison of rupture fronts calculated by different meshes (symmetric or asymmetric) and different numerical fluxes (upwind or mixed); (b) Zoom-in plot of (a); (c) comparison of the cross profile of $Z = 0$ of the shear stress to better visualize the "spatial spikes"; (d) comparison of shear stress time series on the location (5.6, 2.3) km on the fault surface.

Information S1 (Figure S5). Figure S5 in Supporting Information S1 shows that even with a coarse grid, the HFOs caused by the introduction of the central flux are relatively small, especially in the mix flux 2 scheme, and have a smaller impact on the rupture front than the upwind flux because of the reduced dissipation error.

We further compare the propagation of rupture fronts and shear stress time history under the influence of different numerical fluxes and meshes in Figure 7. The results in four cases are compared, namely: symmetric mesh & upwind flux, symmetric mesh & mixed flux, asymmetric mesh & upwind flux, and asymmetric mesh & mixed flux. The contours of the rupture fronts in Figure 7a almost overlap; a zoom-in as in Figure 7b shows the maximum spatial offset of the rupture fronts is only about 30 m. Even for the problematic scheme of asymmetric mesh & upwind flux, its rupture front is close to those of others, because the problem of spatial spikes only occurs after the shear stress is reduced to the dynamic level, hence having little effect on the rupture front itself.

As shown in Figure 7c, we plot the cross profile of $Z = 0$ to compare the SSOs of four cases. The results of symmetric mesh (black and green) are much better than the asymmetric mesh (blue and red) because any numerical errors (dispersion or dissipation) are equal at both sides of the fault surface thus SSOs will not arise. For the asymmetric mesh, the result of mixed flux (red) is much improved compared with those of upwind flux (blue). There are still some minor oscillations caused by the asymmetric mesh (zoom-in plot in Figure 7c) but they will be greatly reduced as the mesh is refined. We also selected one station on the fault surface to compare the shear stress time series of the four cases. Clearly, the application of the mixed flux, even with the asymmetric mesh, successfully converges the stress time series as opposed to the upwind flux method (Figures 7c and 7d). Overall, the fault ruptures calculated by the mixed flux are slightly faster than those calculated by the upwind flux (Figures 7b and 7d). It may be related to the dissipation of the upwind flux, which numerically damps the propagation of the rupture. Nevertheless, the difference between them is insignificant (the maximum time shift in Figure 7d is about 0.02 s).

The geometry of the TPV3 model is relatively simple compared to natural faults with complex fault geometries. By adding two extruded layers, the pairs of elements adjacent to the fault surface are completely symmetric, and the use of the upwind flux is sufficient in this case to prohibit the SSOs. In our experience, even with slightly

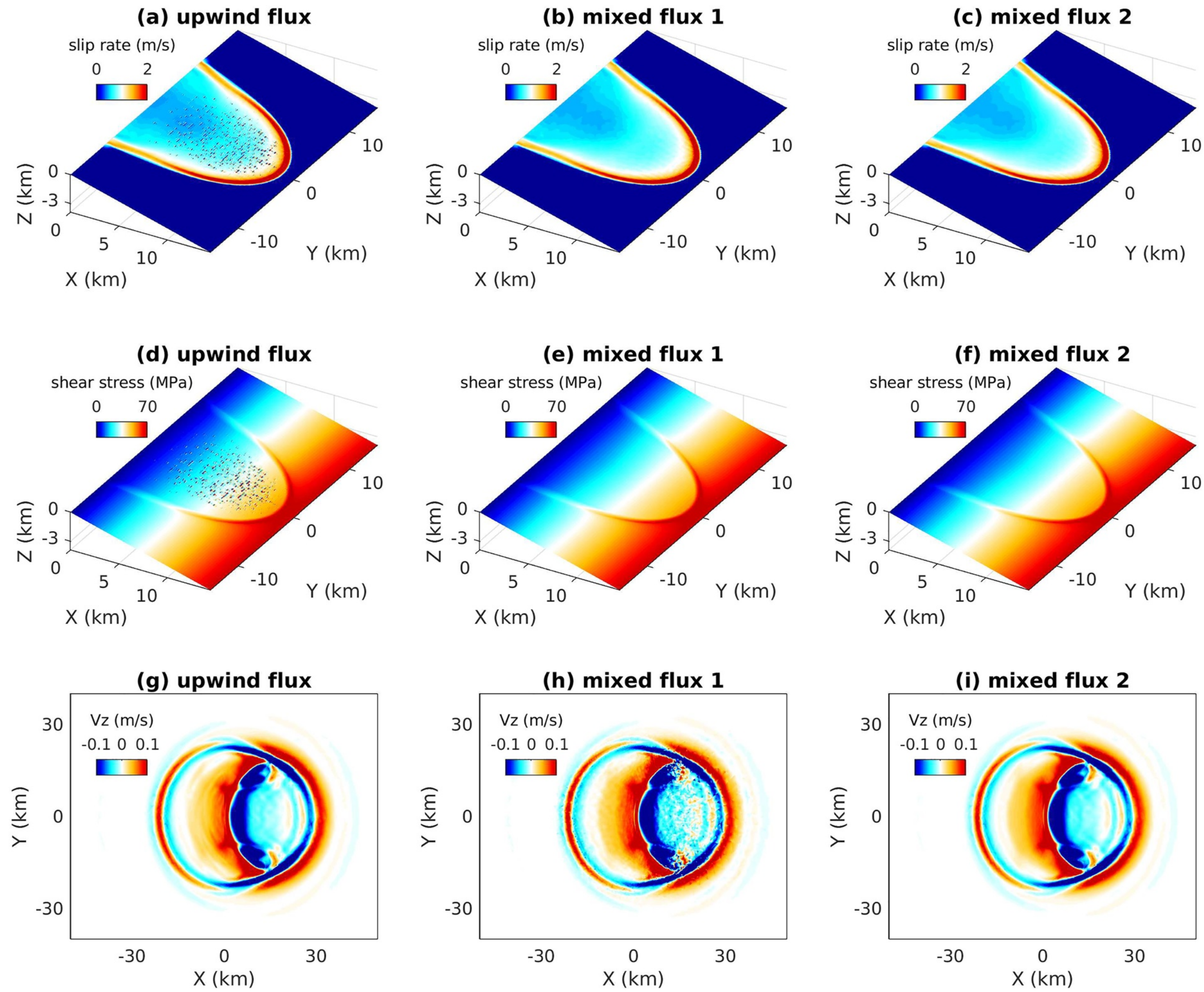


**Figure 8.** Comparison of the trust fault model using different types of numerical fluxes. Left column: upwind flux; middle column: mixed flux 1 (Figure 4a); right column: mixed flux 2 (Figure 4b). First row: snapshots of slip rate at 5 s; second row: snapshots of shear stress at 5 s; third row: snapshots of particle velocity component Vz on the ground surface at 10 s.

asymmetric meshes, maintaining $r_v < 1.5$ can generally produce satisfactory results using upwind flux. However, for more complex fault geometry it is more challenging to generate a nearly symmetric mesh with $r_v < 1.5$. Therefore, the mixed flux method becomes crucial and necessary in scenarios of $r_v > 1.5$, as illustrated in the example below.

The second example is a thrust fault model in half space modified from TPV3 by adding a free surface and modifying the dipping angle to 15°; a shallow dipping angle further exemplifies the mesh asymmetry near the fault surface. Model parameters are listed in Supporting Information S1 (Table S2). Because of the free surface and the low dip angle (15°), it is impossible to generate a symmetric mesh and challenging to generate a nearly symmetric mesh ($r_v < 1.5$). We generated the mesh using Gmsh (Geuzaine & Remacle, 2009). There are 18% elements with $r_v > 1.5$ adjacent to the fault surface (11,624 fault elements in total) and the maximum $r_v$ is ~12 due to the low dip angle. The edge length of the elements near the fault is 300 m and is coarsened to 6 km at the domain boundary. We use the spatial order of $O = 4$ for the simulations.

We first use the upwind flux to simulate the rupture process. As shown in Figures 8a and 8d the SSOs occur in both the slip rate and shear stress snapshots. With the adoption of the mixed flux, the SSOs are completely removed. The snapshots on the fault surface calculated by the two mixed flux approaches (as introduced in Figure 4) are almost identical, indicating that the difference between the two strategies of mixed flux is insignificant for the

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rupture process. The insignificant difference of mixed flux 1 and 2 for the rupture process is also confirmed in a coarse mesh test in Supporting Information S1 (Figures S5a and S5b).

However, as introduced in Section 3.1, the upwind flux scheme is more advantageous than the central flux scheme in suppressing the HFOs. Using the central flux for all the continuous boundaries (Figure 4a) removes the SSOs (compare Figures 8a and 8b, or Figures 8d and 8e), but the HFOs emerges in the far-field ground motion (compare Figures 8g and 8h). This is because the central flux is theoretically non-dissipative (Hesthaven & Warburton, 2008). We found that the central flux at the fault-intersecting faces only is sufficient to suppress the SSOs, whereas upwind flux applied at the rest of the continuous boundaries is effective at suppressing the HFOs (Figure 4b). Therefore, we propose a more advantageous mixed flux approach (mixed flux 2, Figure 4b) which removes the SSOs while suppressing the HFOs. As shown in Figure 8i, the wave field calculated by mixed flux 2 is also free of HFOs, similar to the result of upwind flux (Figure 8g). Therefore, among the three tested numerical fluxes, the scheme of mixed flux 2 should be the best choice for modeling the dynamic rupture process using the DG method with an asymmetric mesh. Note that mixed flux 2 is our default choice for the rest of the modeling cases and henceforth we will refer to our DG method using mixed flux 2 as the “Mixed-Flux-based” DG (MixF-DG) method.

## 4. Numerical Verification

Our proposed mixed flux DG method can be applied to simulate dynamic rupture processes with complex fault geometries, heterogeneous materials, off-fault plasticity, roughness, thermal pressurization, and various versions of fault friction laws. In this section, we benchmark our simulation results with solutions from the *SCEC/USGS Spontaneous Rupture Code Verification Project* (Harris et al., 2009, 2018). In the following, we present the benchmark results for two selected problems and refer the readers to Supporting Information S1 for additional cases.

### 4.1. TPV24: Branching Fault

The first benchmark problem is the SCEC/USGS dynamic rupture model TPV24, which is a branching fault in half space (Harris et al., 2018). The TPV24 model has a vertical planar main fault of 40 km × 20 km and a vertical planar branching fault (12 km × 20 km) that intersects with the main fault at an angel of 30° (Figure S6 in Supporting Information S1). A time-weakening friction law is used in this model. Detailed model parameters can be found on the benchmark website (https://strike.scec.org/cvws) (Harris et al., 2009). The mesh size we used here is 300 m on the fault and gradually increased to 5 km at the domain boundary. We use $O = 4$ for the simulation.

We benchmark our numerical results with those using the modal DG method (ADER-DG) (Pelties et al., 2012), by comparing rupture front contours and synthetic seismograms on the fault, shown in Figures 9a–9c, respectively. Our results are clearly in good agreement with those calculated by the ADER-DG, and the differences between the two methods are within acceptable tolerances of numerical errors (within 1% of relative error of rupture time). This comparison demonstrates that our method can properly simulate dynamic rupture along branching faults.

### 4.2. Rate-State Friction Law With Thermal Pressurization

Our second benchmark example is the TPV105-3D model from the benchmark website, which assumes fault friction follows the strong velocity-weakening rate-state friction law and includes thermal pressurization (TP) as an additional weakening mechanism. We adopt the “pseudo-spectral” method introduced by Noda and Lapusta (2010) to implement the governing equations of temperature and pore pressure changes. The same method is implemented in SeisSol (https://www.scec.org/meetings/2020/am/poster/158).

The TPV105-3D model consists of a vertical strike-slip fault of a size of 44 km by 22 km embedded in a homogeneous half-space (Figure 10a). The fault is governed by the rate and state friction law with flash heating (Di Toro et al., 2004; Rice, 2006) and is further weakened by the reduction of effective normal stress due to the pore pressure increase from shear heating (Andrews, 2002; Noda & Lapusta, 2010). The velocity weakening/strengthening region is set by the spatial heterogeneous friction parameters. The rupture is initiated by a circular region with a radius of 1.5 km where the shear stress exceeds the shear strength. The full description of TPV105-3D can be found in the benchmark verification website https://strike.scec.org/cvws. We use a large computational domain with a size of 200 km by 200 km by 100 km to avoid artificial reflections from domain boundaries. The typical grid size is 200 m near the fault surface and is gradually increased to 20 km at the domain boundary. We perform the numerical simulation using $O = 4$.

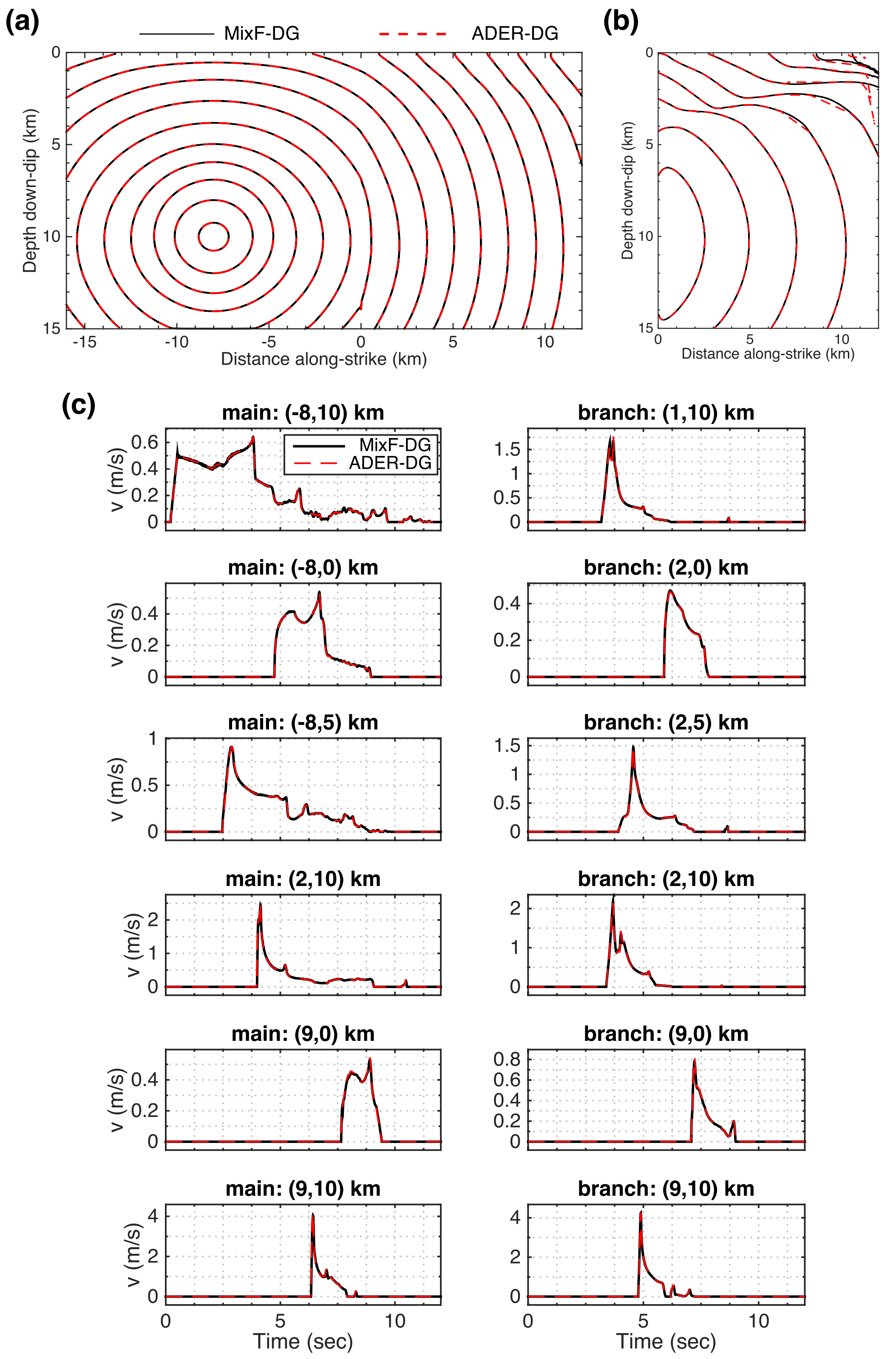


**Figure 9.** (a) Comparison of rupture fronts (every 0.5 s) for the main fault and (b) the branch fault calculated by our method (MixF-DG, solid black line) and the ADER-DG method (Pelties et al., 2012) (dashed red line). (c) Comparison of the slip rate time series at different locations of the main fault plane (left column) and the branch fault plane (right column) for the branch fault model (TPV24).

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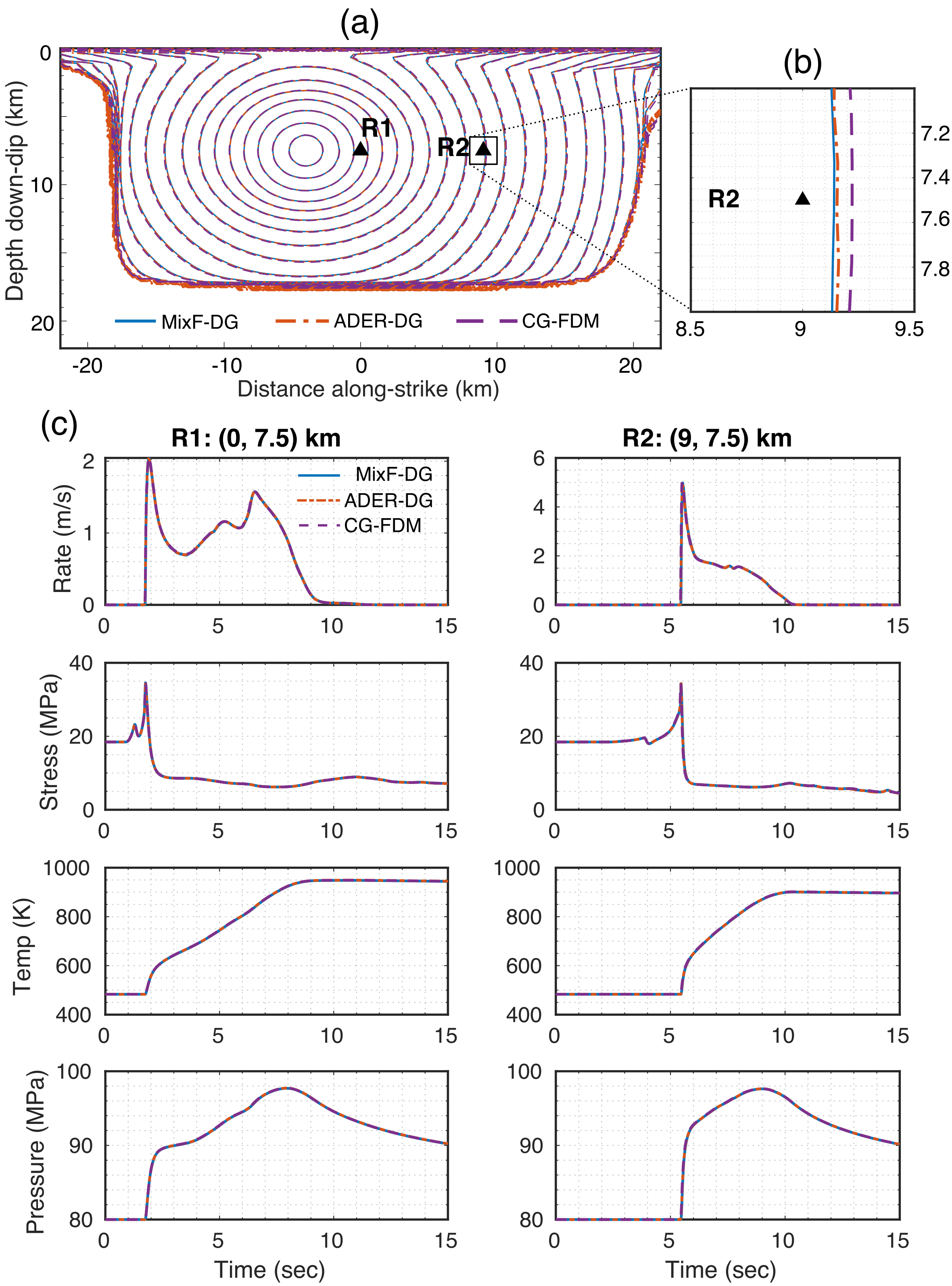


**Figure 10.** (a) Comparison of rupture fronts (every 0.5 s) for the TPV105-3D model calculated by our method (MixF-DG) (solid line), ADER-DG (Pelties et al., 2012) (dash-dotted line) and the GPU-based CG-FDM (W. Zhang et al., 2020) (dashed line); (b) zoomed rupture front contours in (a) near the station R2; (c) Comparison of the time series of the slip rate, shear stress, temperature and pore pressure of the two stations (R1, R2) on the fault surface.

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We compare the rupture front contours and synthetic seismograms on the fault with those from the modal ADER-DG method (Pelties et al., 2012) and a finite difference method CG-FDM (W. Zhang et al., 2020). The typical grid size of the fault in ADER-DG is 250 m and $O = 5$ is used. The grid size of the fault in CG-FDM is 50 m and the spatial order is 4 in the bulk domain and reduced to 2 on the fault. Figure 10 shows that the rupture fronts calculated by the three methods nearly coincide, and the synthetic seismograms are also in good agreement.

To illustrate the effect of thermal pressurization, we also simulate a case identical to TPV105-3D, but without thermal pressurization. As shown in Supporting Information S1 (Figure S7), the introduction of thermal pressurization as an additional weakening mechanism can promote a self-arresting rupture into a runaway rupture with free-surface-induced supershear rupture. This suggests that thermal pressurization is an important factor for ground shaking and seismic hazard assessment.

Readers can also refer to our numerical results uploaded to the benchmark website (https://strike.scec.org/cvws (Harris et al., 2009)) for further direct comparisons.

## 5. Convergence Test

To further evaluate the accuracy of our method, we perform the convergence test as a widely accepted approach for the accuracy analysis of numerical methods to dynamic rupture problems where analytical reference solutions are typically not available (Dalguer & Day, 2007; Day et al., 2005; Pelties et al., 2012; Rojas et al., 2008; Tago et al., 2012; Ye et al., 2020). We measure the numerical error of the method by calculating the relative Root Mean Square (RMS) difference of rupture time, final slip, and the peak slip rate as the mesh is refined (Day et al., 2005). Receivers (stations) distributed every 0.1 km on the fault are used to calculate the RMS difference except for the nucleation patch because of the instantaneous rupture.

To facilitate quantitative comparisons with previous methods, we use the TPV3 model here to test the convergence of our method. The grid size of the mesh of the fault is defined as the side length of the triangle. Here we use grids of 200, 300, 400, 500, 750, 1,000 m, and spatial orders $O = 3$, 4, 5 for convergence tests. The solution of the finest resolution (200 m, O5) is taken as the reference solution. The computational domain of the model is set to be large enough (200 km by 240 km by 200 km) to avoid the effects of artificial reflections.

We use the frontal-Delaunay algorithm of the Gmsh software to generate high-quality equilateral triangles for the fault surface. The grid size at the outer boundaries is set as 10 km, that is, a strong coarsening strategy is used in our case to reduce computation time. In addition, to show the advantage of our proposed mixed-flux method, we generate the mesh without imposing symmetry constraints near the fault even though symmetric mesh can be easily generated for the planar TPV3 model. Therefore, the strongly coarsened mesh we use here is highly asymmetric near the fault (Figure 3b).

We compare the convergence results of our mixed-flux-based nodal discontinuous Galerkin (MixF-DG) method with those of other methods, which are summarized in Table S3 of the Supporting Information S1, Table 1 and visualized in Figure 11. We choose the finite-different method (DFM) (Day et al., 2005), the boundary integral method (BI) (Day et al., 2005), the upwind-flux-based modal discontinuous Galerkin method (ADER-DG) (Pelties et al., 2012) and the central-flux-based nodal Galerkin method (DGCrack) (Tago et al., 2012) for comparison. For a quantitative and fair comparison of the three DG methods, we need to unify the definition of the grid size due to different meshing strategies. The meshes of DGCrack and our method are both generated by Gmsh, and the fault grid size is naturally defined as the side length of the equilateral triangle on the fault. However, the fault mesh of ADER-DG is first generated as a quadrilateral and then divided into two triangles. They define the mesh size as the length of the longest side of a triangle ($\sqrt{2}$ times the length of the shortest side), which is different from the definition of grid size in DGCrack and our method. Choosing either the shortest side or the longest side will result in an unfair comparison. Therefore, we define the equivalent grid size as the side length of an equal-area equilateral triangle. Using the unified grid size definition, the number of fault meshes of the three methods keeps nearly equal for the same grid size.

Figure 11a shows the convergence results of the rupture time. The RMS error and the measured convergence rate (~3) of our method (MixF-DG) and ADER-DG are close (also see Table 1). The convergence rate saturates at the order of 3 and does not increase further, which is in line with theoretical expectations (Godunov & Bohachevsky, 1959; Hesthaven & Warburton, 2008) and previous numerical works (Dalguer & Day, 2007; Day

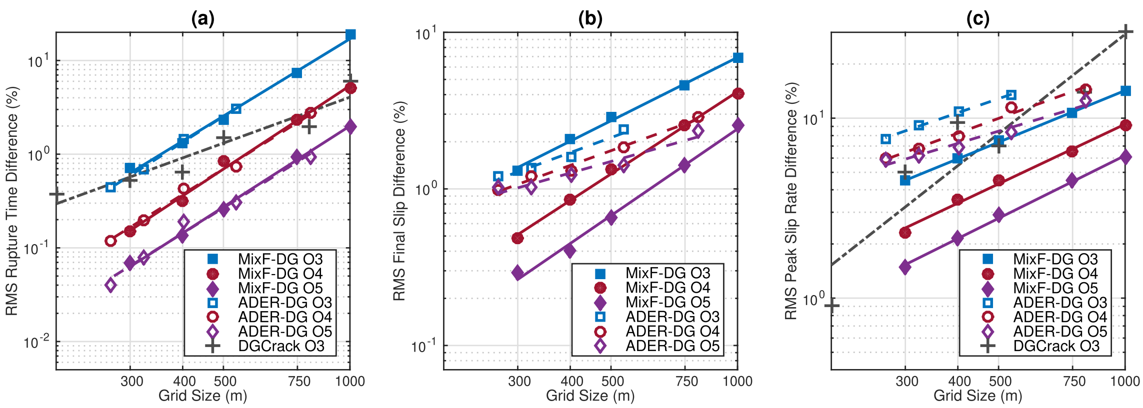


**Figure 11.** Convergence results for the TPV3 test case using our MixF-DG, ADER-DG (Pelties et al., 2012) and DGCrack (Tago et al., 2012) method. The markers represent the RMS error calculated by different order-of-accuracy $O = 3, 4, 5$. The lines with the same color represent the regression of corresponding markers, which yields the measured convergence rate exponents in Table 1.

et al., 2005; Pelties et al., 2012; Rojas et al., 2008; Tago et al., 2012; Ye et al., 2020) because of the existence of non-smoothness. However, even if the increase of $O$ does not increase the convergence rate, the reduction of the magnitude of error is significant. For example, the three cases of 500 m O5, 400 m O4, and 300 m O3 have the same resolution (100 m), but the case of 500 m O5 has the smallest error. This result suggests that the accuracy of dynamic rupture solutions can be better improved by using higher-order methods than by refining the mesh. Among the three DG methods, the convergence rate of DGCrack (1.65) is significantly lower than our method and ADER-DG (~3, see Table 1). However, in the case of the coarse grid (1,000 m O3), DGCrack has the best performance whereas no physical solution is obtained with the ADER-DG method (see Pelties et al., 2012). Our method performs well on both coarse and fine meshes.

Figure 11b shows the convergence test of the final slip. When $O = 3$, our MixF-DG method is slightly worse than the ADER-DG method. However, our method converges rapidly with the increase of grid resolution and $O$. The final slip results by the ADER-DG method have the lowest convergence rate (<1), while other methods converge faster with a rate over 1 (Table 1). Our method achieves the best performance in terms of the convergence rate (1.84) of the final slip in Table 1.

Figure 11c shows the convergence test of the peak slip rate, where our method significantly outperforms the ADER-DG method in any case in terms of the RMS magnitude. However, in terms of the convergence rate, our method is only slightly better than the ADER-DG method. The convergence rate of the peak slip rate is relatively low compared to the rupture time or final slip (Table 1). An exception is the DGCrack method, which achieves the highest converge rate of 1.83 among these methods in Table 1. However, its convergence stability is worse than other methods, and its measurement results are relatively oscillating near the reference convergence line (Figure 11c).

**Table 1**
*Comparison of TPV3 Convergence Rate Exponents for Different Methods*

| Method | Rupture time | Final slip | Peak slip rate |
|---|---|---|---|
| MixF-DG O3 | 2.74 | 1.35 | 0.95 |
| MixF-DG O4 | 2.96 | 1.75 | 1.10 |
| MixF-DG O5 | 2.87 | 1.84 | 1.16 |
| ADER-DG O3[a] | 2.84 | 0.99 | 0.80 |
| ADER-DG O4[a] | 2.83 | 0.97 | 0.85 |
| ADER-DG O5[a] | 2.83 | 0.75 | 0.70 |
| DGCrack[b] | 1.65 | 1.52 | 1.83 |
| DFM[c] | 2.96 | 1.58 | 1.18 |
| BI[c] | 2.74 | 1.53 | 1.19 |

[a]Pelties et al. (2012). [b]Tago et al. (2012). [c]Day et al. (2005).

Overall, our mixed-flux DG method performs well and is competitive with the state-of-the-art methods in convergence tests. In all tests, we did not impose symmetry constraints on the mesh and adopted strong coarsening to reduce computation time. The good and stable convergence performance despite the use of asymmetric mesh shows the robust accuracy and meshing flexibility of our method.

## 6. 2008 Wenchuan Earthquake Rupture Simulation

In this section, we demonstrate that the mixed-flux DG method can be applied to model earthquake ruptures on natural faults of complex geometry using the 2008 Mw 7.9 Wenchuan earthquake as an example. Our purpose here is to illustrate our newly developed method can properly simulate the dynamic rupture process given the fault geometry and material properties constrained by previous studies; we do not intend in this work to use the

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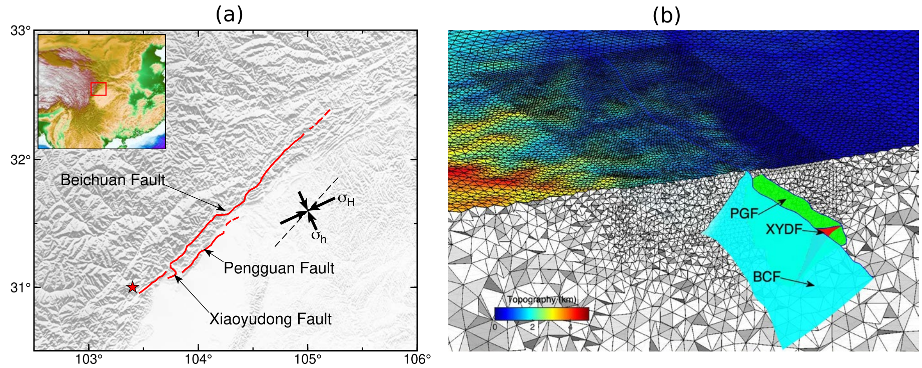


**Figure 12.** (a) The surface trace of the 2008 Wenchuan earthquake fault rupture. The ruptured faults include the Beichuan fault (BCF), the Pengguan fault (PGF) and the Xiaoyudong fault (XYDF). The *Y*-axis (dashed black line) of the computational coordinate system is approximately NE 45°. The triaxial stress is assumed to be uniform in the horizontal direction. The black thick arrows indicate the direction and relative magnitude of the maximum/minimum horizontal principal stress ($\sigma_H$ and $\sigma_h$). (b) Close-up and cross-sectional views of the unstructured tetrahedral mesh of the Wenchuan earthquake model. The colorbar indicates surface topography. The edge length of the elements on the fault is 2 km and coarsened to 10 km at the domain boundary. The mesh contains a total of 1,079,239 tetrahedrons.

simulated results to interpret specific Wenchuan earthquake rupture mechanisms. Therefore here we do not make quantitative comparisons between our simulation results and observations. Rather, we will focus on the qualitative comparison and highlight the first-order importance of fault geometry.

The Wenchuan earthquake occurred in the Longmenshan thrust belt on the eastern margin of the Tibetan Plateau. Field surveys, GPS and InSAR measurements, and kinematic inversions of the rupture process show that the slip distribution of the Wenchuan earthquake is highly heterogeneous (Shen et al., 2009; Wan et al., 2016; Xu et al., 2009). For example, the southwestern part of the Beichuan fault presents thrust and right-lateral strike-slip, while its northeastern part is dominated by strike-slip. Numerical simulations have been conducted (Duan, 2010; Z. Zhang et al., 2019) to explain the observations and understand the underlying physics of the rupture process. However, the fault geometry is usually simplified due to the numerical difficulty of incorporating multiple fault segments. For example, the multi-fault system has been to the first order approximated as a planar dipping fault, and the non-uniform slip distribution is explained by varying the background stress tensor along the fault strike (Duan, 2010). Alternatively, Z. Zhang et al. (2019) showed that the non-uniform slip distribution can also be reproduced with non-planar fault geometry under uniform background stress. A recent study by Tang et al. (2021) incorporated multi-fault segments and can produce simulation results (e.g., final coseismic slip distribution, surface displacement) in good agreement with GPS and InSAR measurements. However, due to the limited meshing and modeling flexibility of their numerical method (redevelopment based on the commercial software ABAQUS/Explicit), Tang et al. (2021) excluded the Xiaoyudong fault segment, which is estimated to have experienced up to about 3 m of coseismic slip (Tan et al., 2012). Chen et al. (2013) suggests that the Xiaoyudong rupture is not a passive tear fault but an active participator of slip partitioning on multiple faults within the Longmenshan thrust system. Therefore, we include the Xiaoyudong fault in the model (Figure 12a).

The 3D fault geometry is constructed based on observations from previous studies (Hubbard et al., 2010; Shen et al., 2009; Wan et al., 2016), including the Beichuan fault, the Pengguan fault, and the Xiaoyudong fault segments. We also include the surface topography effect during the simulation to account for the large differences in terrain heights (~4 km) in this region (Figure 12). As shown in Figure 12b, we utilize an external mesh generation software CUBIT (http://cubit.sandia.gov/) to automatically generate tetrahedral meshes that include both topography and multi-fault geometry and label the free surfaces and fault surfaces as boundary conditions. The rest of the boundaries are treated as interior or absorbing boundaries and are handled automatically by our program (Section 2.2). To focus on the fault geometry effect, we set a homogeneous medium (Table S4 in Supporting Information S1). A linear slip-weakening friction law is adopted in the dynamic rupture simulation. The background stress field is horizontally uniform and varies only with depth. The friction and stress parameters are basically adopted from previous modeling works (Tang et al., 2021; Z. Zhang et al., 2019). As shown in Figure 12b, we utilize an external mesh generation software CUBIT (http://cubit.sandia.gov/) to automatically generate tetrahedral meshes that include both topography and multi-fault geometry and label the free surfaces and

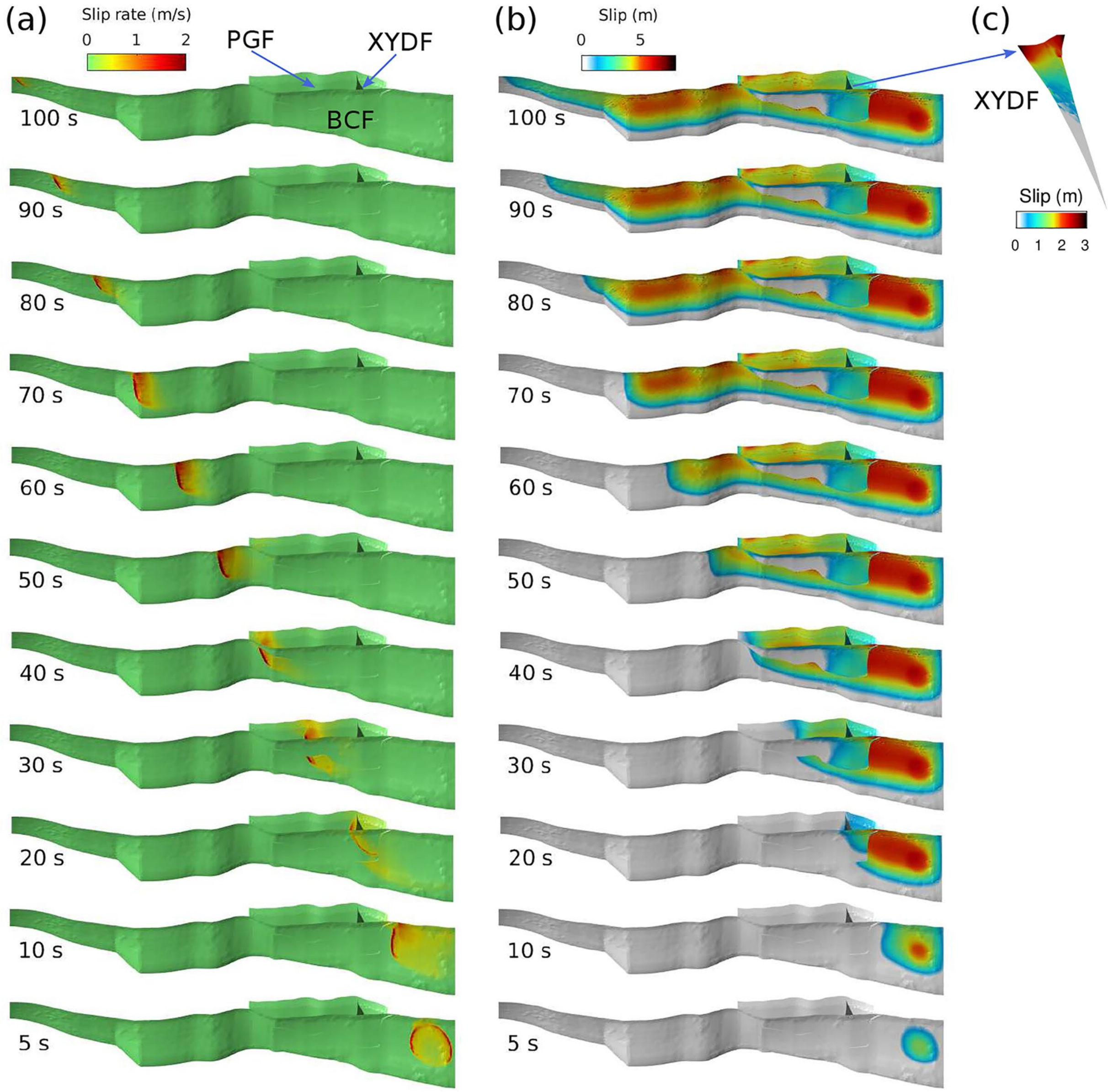


**Figure 13.** Snapshots of (a) slip rate and (b) cumulative slip for the dynamic rupture process of the 2008 Mw 7.9 Wenchuan earthquake. (c) The zoom-in plot of the final slip of the Xiaoyudong fault.

fault surfaces as boundary conditions. The rest of the boundaries are treated as interior or absorbing boundaries and are handled automatically by our program (Section 2.2). To focus on the fault geometry effect, we set a homogeneous medium (Table S4 in Supporting Information S1). A linear slip-weakening friction law is adopted in the dynamic rupture simulation. The background stress field is horizontally uniform and varies only with depth. The friction and stress parameters are basically adopted from previous modeling works (Tang et al., 2021; Z. Zhang et al., 2019). We set the initial stress to be 5% above the peak stress at a circular patch of radius 8 km at the depth of 19 km to nucleate the rupture from the southwest end of the Beichuan fault. Detailed model parameters are listed in Supporting Information S1 (Table S4). We used 256 CPU processors in Compute Canada (Beluga) for parallel computing and the total computing time is about 90 min.

Figure 13 shows the snapshots of simulated fault slip rate (left) and cumulative slip (right). Between 0 and 10 s, the rupture nucleated at the southwest end of the Beichuan fault and propagated toward the northeast. By the time of about 20 s, the southwest segment of the Beichuan fault accumulated up to 8 m of slip and remained the segment of the highest coseismic slip. At the same time, rupture jumped across to the Xiaoyudong fault and the Pengguan fault resulting in ∼3 m of coseismic slip, while continuing to propagate along the deep part of Beichuan fault with slightly less amount of coseismic slip (Figure 13a, 20–30 s). At about 40 s, the rupture ran to the northeast end of the Pengguan fault, where it re-converged with the rupture on the Beichuan fault and continued to rupture further to the northeast. The final coseismic slip is highly heterogeneous, with the maximum slip accumulated near the nucleation section at the southwest segment of the Beichuan fault, which is consistent with the results of kinematic inversion (Shen et al., 2009; Wan et al., 2016). The Xiaoyudong fault was not included in previous numerical modeling works of the Wenchuan earthquake rupture due to the limitation of numerical methods (Tang et al., 2021; Z. Zhang et al., 2019). Here we demonstrate that its inclusion results in a coseismic slip of ∼3 m (Figure 13c), roughly consistent with field investigations (Tan et al., 2012). Because we did not rotate the direction of the principal stress field along strike (as was done in Duan (2010)) and assumed a homogeneous medium in our simulation, the non-uniform coseismic slip distribution is attributed to the complex fault geometry, highlighting its first order importance.

## 7. Discussions

To deal with the complex fault geometry of many natural earthquakes (e.g., the 2008 Mw 7.9 Wenchuan earthquake), we developed a nodal discontinuous Galerkin method for 3D dynamic rupture modeling. We have modified the upwind flux scheme (mixed flux) for reliability and tested parallel computing and mesh coarsening techniques for computational efficiency. The mixed flux scheme is important to achieve a stable simulation when asymmetric mesh near the fault is used. Below we discuss the underlying reasons why mixed flux is suitable for dynamic rupture problems and their possible implications.

### 7.1. Implications of the Mixed Flux Scheme

The choice of the numerical flux is essential to the numerical algorithms implemented with the discontinuous Galerkin method. For example, the upwind flux is universally used to deal with all boundary conditions for wave propagation in isotropic, anisotropic, poroelastic and acoustic-elastic media (Wilcox et al., 2010; Zhan et al., 2020). While universal adoption of a single kind of numerical flux is typically sufficient for most DG-based models with multiple types of boundary conditions, there do exist some, albeit few, studies using more than one type of numerical flux for various purposes. For example, He, Yang, Ma, and Qiu (2020) uses the linear combination of Godunov flux and central flux (called "modified flux") to improve the performance of elastic wave modeling in isotropic and anisotropic media. The numerical dispersion errors are reduced thereby increasing accuracy after using the modified flux. In our work for the dynamic rupture model, the use of mixed flux is crucial as the instability problem (SSOs) occurs with asymmetric mesh when only the upwind flux is applied. The choice of a numerical flux suitable for specific model problems and meshes is paramount to the numerical stability of the simulation.

Although our work has proved through numerical tests that mixed-flux produces more stable results than upwind-flux in dynamic rupture models with asymmetric meshes, the underlying mathematical principles are still not fully understood. According to the analysis of Section 3 (Figure 5), the idea of mixed-flux should not be limited to the tetrahedral mesh only. Hence it is of interest to further verify whether mixed-flux still works in different types of meshes. For this purpose, we developed a dynamic rupture program based on hexahedral meshes following the method of Persson (2013) (see Text S9 in Supporting Information S1). Using the upwind-flux-based DG method with a highly asymmetric hexahedral mesh also results in the problem of spatial spikes while the mixed-flux-based method does not. The fact that mixed flux scheme is effective in inhibiting spatial spikes regardless of the mesh type further supports the robustness of our new method, which may also be useful for other types of physical problems with similar mixed continuous-discontinuous boundaries.

### 7.2. Stability

Previous work has demonstrated the stability of numerical methods for seismic wave and dynamic rupture modeling based on upwind flux (de la Puente et al., 2009; Duru et al., 2021; Kaneko et al., 2008; Pelties et al., 2012; Wilcox et al., 2010) or central flux (Delcourte et al., 2009; Tago et al., 2012). Our mixed-flux method is as stable as the upwind flux and the central flux as it is a mixture use of them at different types of boundaries. By analyzing the discrete kinematic and strain energy of the whole computational domain, we confirmed in a stability test (Text S5 and Figure S2 in Supporting Information S1) that our method is stable even when the simulation time is 10

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times longer than the rupture process. The long-time simulation stability is critical for the strong ground motion modeling used in seismic hazard evaluation.

### 7.3. Parallelization and Scalability

The discontinuous Galerkin method has clearly more degrees of freedom than the traditional finite element method for the same mesh (see Figures 1b and 1c). For a single-process program, the DG method is less computationally efficient than the traditional FEM. However, the DG method is much more adaptive to parallel multi-thread computation than the traditional FEMs, because the linear system of equations is solved locally instead of globally for the traditional FEM. The scaling test of MPI parallel computing in Figure S3 of the Supporting Information S1 shows the measured (as opposed to idealized) per-step run-time scales with the number of processors at nearly the ideal scaling from 32 to 256 processors (with the order-of-accuracy $O = 2, 4, 6$). The scaling slightly deviates from the ideal scaling at 512 processors, mostly due to the increase in message passing time for a large number of processors. At 512 processors, the highest order ($O = 6$) simulation performs better than low-order tests in scalability because of the increased computational time for the matrix multiplication (Equation S19 in Supporting Information S1). We expect the good parallel scaling to extend to an even larger number of processors at a relatively high order of simulations.

### 7.4. Effects of Mesh Coarsening

One of the most important advantages of tetrahedral mesh is its flexibility to allow a great degree of mesh coarsening, which significantly reduces the number of mesh elements thereby the computation time. A detailed comparison of a 2D modal DG method (ADER-DG) is given in de la Puente et al. (2009), demonstrating that the mesh coarsening has little effect on the rupture process. Here we provide a quantitative investigation (Figure S4 in Supporting Information S1) of the effects of mesh coarsening for the information of future users of our method. We found that even a simple coarsening strategy (e.g., linear or quadratic) would not affect the rupture process much. If we keep the 5 equal-size layers around the fault surface, the effect on the rupture process is invisible even if a very strong mesh coarsening is applied (e.g., the mesh size at the far-field domain boundaries is 50 times that of the fault). Obviously, mesh coarsening has a relatively large effect on the far-field seismogram (Figure S4d in Supporting Information S1) because of the coarse mesh near those stations. Overall, the comparison in Figure S4 of the Supporting Information S1 suggests that even a very strong mesh coarsening will not greatly affect the results, although caution is needed in far-field high-frequency waveform interpretations.

### 7.5. Limitations and Future Works

Absorbing boundary treatment is very important for volumetric numerical methods such as FDM, FVM, and FEM. In all the benchmark problems and the Wenchuan earthquake simulation case discussed above, we adopted the first-order absorbing boundary condition based on the numerical flux (Equation 9). More advanced absorbing boundary treatment such as the perfectly matched layer (PML) (Berenger, 1994), can be implemented in the current framework. However, thanks to the strong coarsening ability of the tetrahedral mesh, the use of PML became less urgent as we can accommodate a larger computational area without significantly increasing the number of mesh elements. By comparing the results with those of other methods, it can be shown that the fault rupture process we calculated is basically not affected by the spurious reflected waves. However, for some special cases, for example, when the aspect ratio of the computational domain is very large, the near-grazing incident waves (W. Zhang & Shen, 2010) are difficult to absorb with the first-order absorbing boundary condition. In this case, it is necessary to use the PML technique to deal with spurious reflected waves.

The method is currently limited to the first-order mesh, that is, the straight-edge mesh. The curved fault surface is represented by the piecewise linear segments. Using straight-edged elements will inevitably result in a kink-like geometry of adjacent elements. This kink-like geometry creates stress concentrations that cause the stress snapshot to appear as a small amplitude spatial oscillation. Refining the grid can make this kink smaller at the expense of increased computational load. One of the natural adjustments is to use a curved mesh DG method. However, the computational and memory costs will increase due to the integration of curvilinear elements in tetrahedral mesh, especially when the boundary geometry is represented by very high-order polynomials in high dimensions. Future work needs to devote to the efficient implementation of the DG method based on curved meshes. With the

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burgeoning computational resources, extensions to more powerful platforms (e.g., cloud computing, GPUs and exascale computing) should also be encouraged (Ramos et al., 2022).

A natural next step is to apply our method to simulate real-world earthquake ruptures in complex fault systems. In this work, we focused on the geometric control of the Wenchuan earthquake rupture process, while keeping the loading stress and material properties uniform. The heterogeneous distribution of these conditions do need to be taken into consideration in order to fully understand the details of the Wenchuan earthquake rupture (Duan, 2010; Wan et al., 2016). The enhanced flexibility of our mixed-flux method facilitates dynamic rupture simulations of such complex earthquake ruptures. For example, the fault geometry of the 2019 Mw 6.4/7.1 Ridgecrest earthquakes is highly complex with intersections, step-over, and branching structures (Ross et al., 2019), which may lead to complex interactions between different faults during the rupture process. The recent 2023 Mw 7.8 and 7.6 Turkey-Syria earthquake doublet involved nucleation on a small branch fault, rupture jump to a nearby main fault, possible supershear rupture (Melgar et al., 2023; Rosakis et al., 2023). Our newly developed method can properly handle the geometric and material heterogeneities of such complex rupture processes to better understand their physical mechanisms.

## 8. Conclusion

We developed a new nodal discontinuous Galerkin method to model 3D dynamic rupture problems with a fully unstructured tetrahedral mesh. A heterogeneous upwind flux scheme for the fault discontinuity boundary condition is derived based on the velocity-strain elastodynamic equation. We found that in cases of fault-bounding asymmetric mesh, the universal adoption of upwind flux for all boundary conditions will lead to spatial oscillations, especially in the stress components. To circumvent this problem, we proposed a new mixed flux scheme, which applies central flux only to the surfaces immediately adjacent to the fault and uses upwind flux for all other surfaces. The mixed flux scheme subtly removes the instability of spatial oscillations without losing accuracy or increasing computational burden. Our results suggested that the effectiveness of our proposed mixed central/upwind flux for dynamic rupture problems is related to the inherent numerical dissipation, but a more rigorous, mathematical analysis deserves further studies.

Convergence and stability tests for the mixed-flux scheme are carried out, showing that our method has high accuracy and strong stability. The use of numerical fluxes in the DG scheme enables an explicit time integration scheme, hence the easy implementation of large-scale parallel computation. Our program shows satisfactory parallel efficiency up to ~500 CPU processors using Message Passing Interface (MPI), with the potential for optimization and scaling up on supercomputers. We verified the simulation results from the mixed-flux method against published results from a broad range of SCEC/USGS dynamic rupture benchmark problems, including bimaterial faults, off-fault plasticity, thermal pressurization, complex fault geometries, and various forms of friction laws. We conclude that our work provides a reliable, flexible, and efficient tool to model dynamic rupture processes for complex fault geometries and heterogeneous material properties.

We applied our method to simulate the dynamic rupture process of the 2008 Wenchuan earthquake, including multiple fault branches, step-overs, a previously neglected sub-parallel fault (Xiaoyudong), as well as realistic topography. Despite the simple adoption of uniform background stress and media properties, our first-order model results (rupture propagation speed, coseismic slip distribution) are consistent with previous field observations and modeling studies (Shen et al., 2009; Xu et al., 2009; W. Zhang et al., 2020). Our simulation results also show that coseismic slip can be up to 3 m on the Xiaoyudong fault, consistent with post-Wenchuan geological surveys, and allows us to investigate in detail the interaction between multi-fault segments. Our new algorithm combines the flexibility of the nodal discontinuous Galerkin method to represent complex fault geometries, heterogeneous material properties, and high tolerance of the mixed-flux approach for mesh quality, thus offering an efficient numerical approach to systematically explore the influences of fault zone properties on earthquake rupture dynamics.

## Data Availability Statement

The dynamic rupture simulation software DRDG3D developed in this work has been open-sourced on GitHub: https://github.com/wqseis/drdg3d and archived on Zenodo: https://doi.org/10.5281/zenodo.8000437. The results of the benchmark problems produced by our method are publicly available from the website of the *SCEC/USGS*

*Spontaneous Rupture Code Verification Project*: https://strike.scec.org/cvws (Harris et al., 2009). The data of other cases can be downloaded from Zenodo: https://doi.org/10.5281/zenodo.7758503.

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## Erratum

The originally published version of this article inadvertently omitted thanking the editor, an associate editor, and two anonymous reviewers. The first sentence of the Acknowledgments should read: "We appreciate the constructive comments and suggestions from editor Rachel Abercrombie, an associate editor, and two anonymous reviewers during the review process." In addition, the last two sentences of the Acknowledgments, beginning with "All simulations were run" and ending with "Digital Research Alliance of Canada (ID 2601)," should read: "All simulations in this work were performed on HPC Béluga and Narval of Digital Research Alliance of Canada via a Research Allocation Competition award to Y. Liu at McGill. W. Zhang and Y. Liu acknowledge the support of the Natural Sciences and Engineering Research Council of Canada (NSERC): Discovery Grant RGPAS/522498-2018, Wares Science Innovation Prospectors Fund 2020. X. Chen acknowledges the support of the National Natural Science Foundation of China (Grant U1901602) and Guangdong Provincial Key Laboratory of Geophysical High-resolution Imaging Technology (2022B1212010002)." The errors have been corrected, and this version may be considered the authoritative version of record.